Macro Finance Seminar Notes: Lars Peter Hansen

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"Asset pricing, decision theory, and public policy design"

See Souganidis paper, ask Ethan about viscosity solutions, ask Anne about mean-field games.

"Intertemporal marginal social valuations" [SJ: ?] (exs: social cost of carbon, social cost of R&D)

Math trick: Girsanov

App: big project R&D motivated by climate change (climate economics, environmental economics, ask Tucker and AB)

Formulation is nontrivial; and "model interpretation", not "model solution"

"We confront broad-based uncertainties arising from imperfect measurement and quantification"

The NASEM released a 2017 recommendation on the social cost of carbon, with four distinct modules: the specialists subdivide, passing evaluations from one to the next. E.g., socioeconomic $\to$ climate $\to$ damage $\to$ discounting

But this is lacking an "integrated science" approach...

Game plan: build a value function; form "stochastic impulse responses"; deduce pricing formula; construct an uncertainty-averse tilted probability measure; model big changes as Poissonian (small shocks are modeled as Brownian increments); deduce "additive decompositions" of the contributions to the marginal valuations.

The state evolves according to:

$dX_t = \mu(X_t)dt + \sigma(X_t)dW_t$ (drop the $(\cdot)$'s going forward)

Construct a "stochastic IR" to measure the impact of a "marginal change in a state"

We want to leverage the *theory of stochastic flows* by Kunita. [SJ: Kunita of Kunita–Watanabe fame? See my stochastic calculus notes.] Perturb the initial condition X_0. The stochastic IR process Λ captures the effect on future X's. [SJ: Formally, "$\Lambda := dX/dX_0$"?]

Component i of the IR, denoted Λ^i, evolves as

$d\Lambda_t^i = (\Lambda_t)'\partial_x \mu_i dt + (\Lambda_t)'\partial_x \sigma_i dW_t$ (basically, chain rule)

[SJ: Derive this for fun, learn stochastic flows.]

Stack all the Λ^i's and study joint dynamics of (Λ, X)

Linear VAR counterpart is a special case ($\mu(x) = cx, \sigma(x) \equiv \sigma$); in this case, Λ is deterministic. [SJ: Derive this for fun.]

[SJ: Q from audience: which components of X are endogenous vs exogenous? Reply: handle it later.]

Now we construct a value function associated with a disconuted objective with discount rate δ and utility U. There are maximizing controls (investments and emissions) and a minimizing probability measure.

Initial pricing formula: partial derivatives as asset prices

$\partial_x V(X_0) \cdot \Lambda_0 = \delta \int_0^\infty \exp(-\delta t)\mathbb{E}_0[\partial_x U(X_t) \cdot \Lambda_t]dt$

[SJ: How to read this equation? Summing the contributions of "marginal utility contributions in the future"; this is the term $\partial_x U(X_t) \cdot \Lambda_t$.]

[SJ: Q from Sean about changing coordinates to simplify Λ. Reply unclear to me.]

How does this differ from a co-state formulation of optimal control? No optimization ("envelope condition"), vector of interconnected co-state equations to solve forward, etc... [SJ: ?] This is easier to interpret.

State-dependent policies: $\mu(x) = \hat{\mu}[x, \Gamma(x)]$ and $U(x) = \hat{U}[x, \Gamma(x)]$, where Γ is the "policy rule". (Here, we're just making explicit the dependence of μ, U on the policy rule.)

The marginal values are $\delta\partial_\gamma \hat{U} \cdot \Delta$ and $\partial_x V \cdot [\partial_{\gamma^\top} \hat{\mu}\Delta]$. [SJ: ?]

Adding in jumps: imagine there are L kinds of jumps, with different Poisson process intensities. Let V^ℓ denote the "continuation value function" for the first jump being of type ℓ, conditional expectation.

Jump contributions. [SJ: Add picture. It would've been helpful for him to label each term in these chain rule equations. As Sean clarified, this term from time t (which itself is a sum over possible jumps $\ell \in [L]$) gets discounted back to time zero via $\delta \exp(-\delta t)$, then integrated over time $t \in [0, +\infty)$, as in the marginal value formula before.]

Part II: adjusting valuation for uncertainty.

The term "deep uncertainty" shows up in the literature, due to the complexity of climate change forecasting. Borrow insights from decision theory.

Economists, when consulted by policymakers, are often afraid to express uncertainty, because

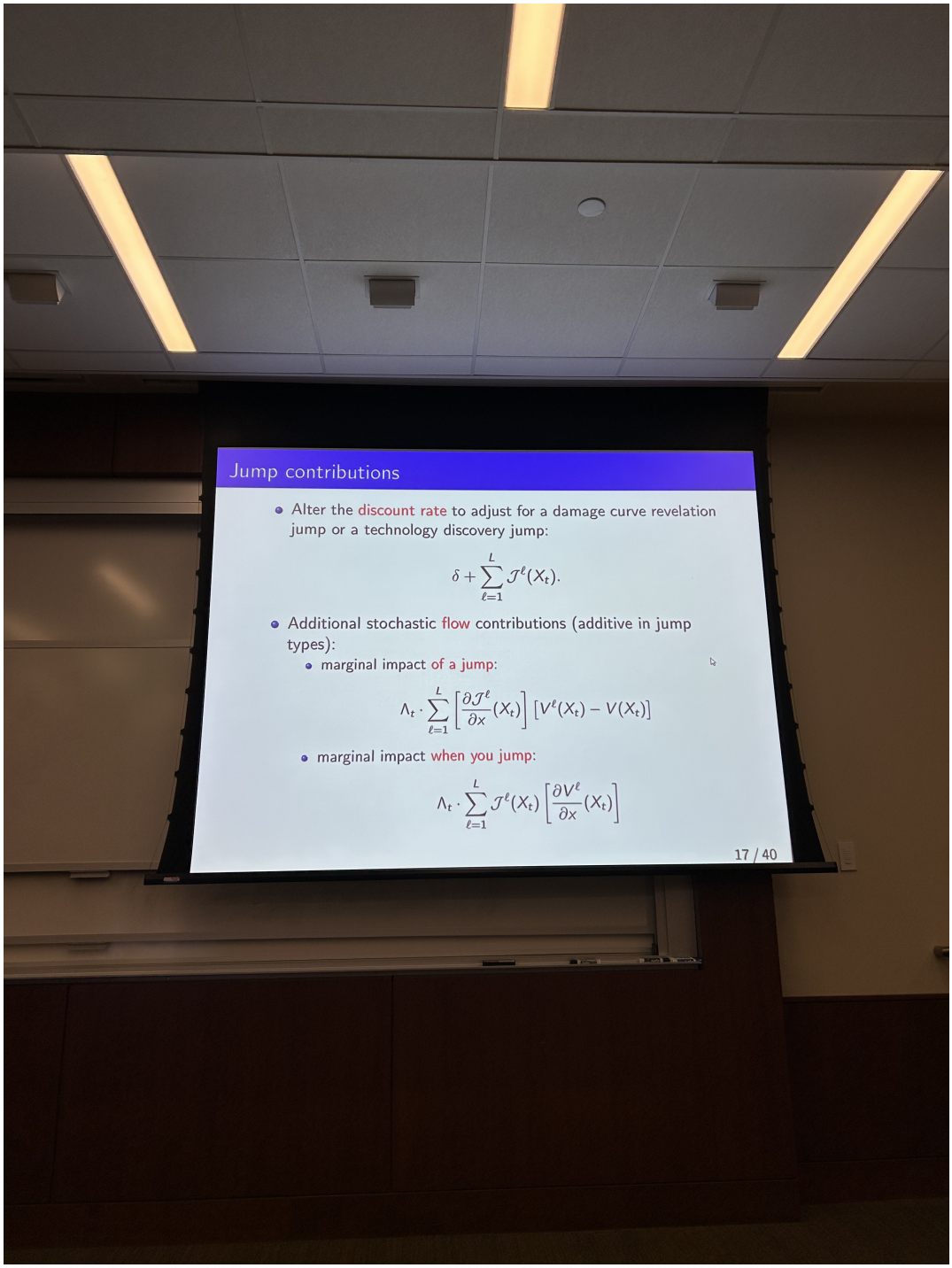


Figure 1: Formula for marginal contributions of jumps at time t.

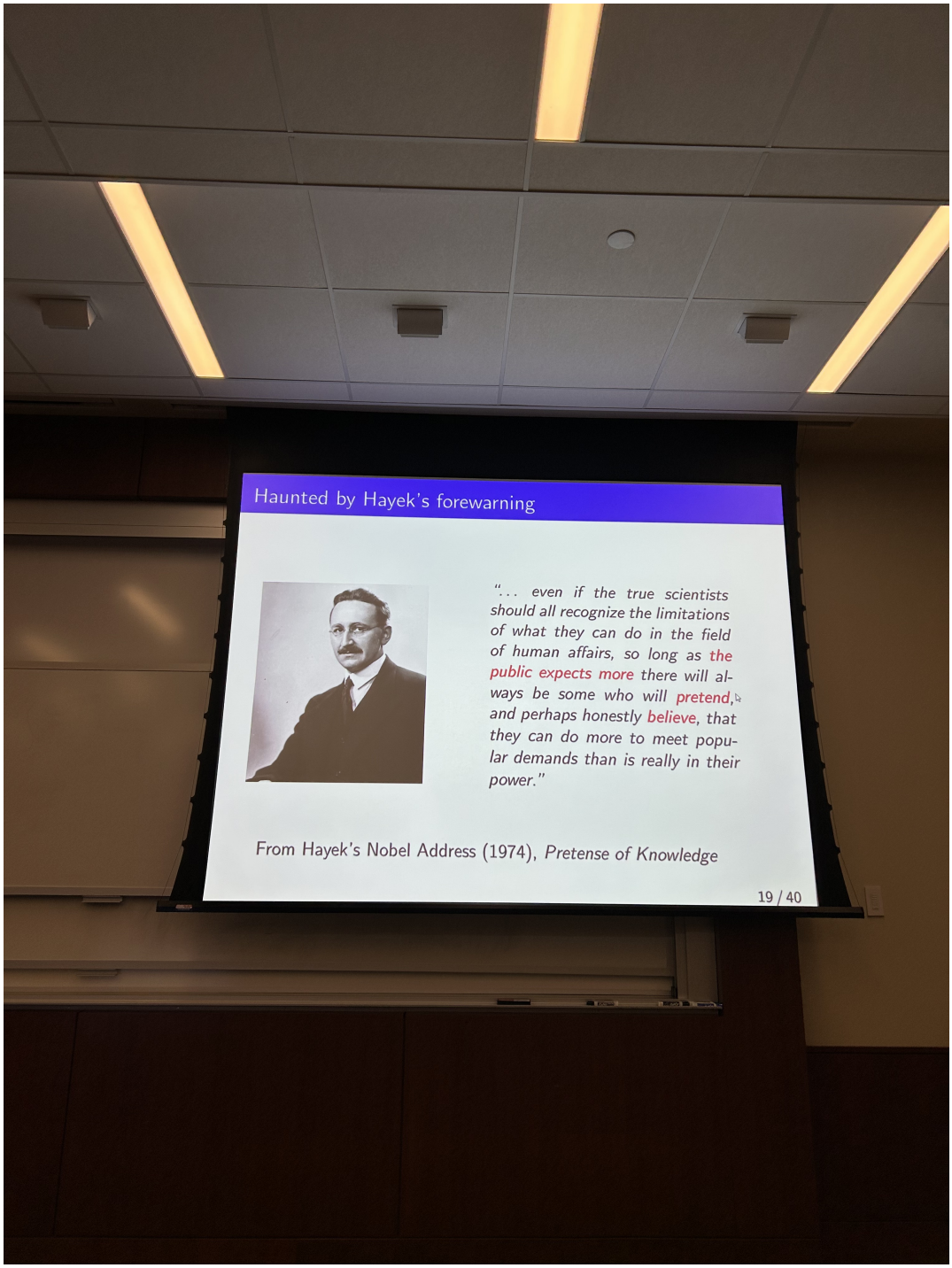


Figure 2: Is economics even a science? In this essay, Hayek roasts econometrics.

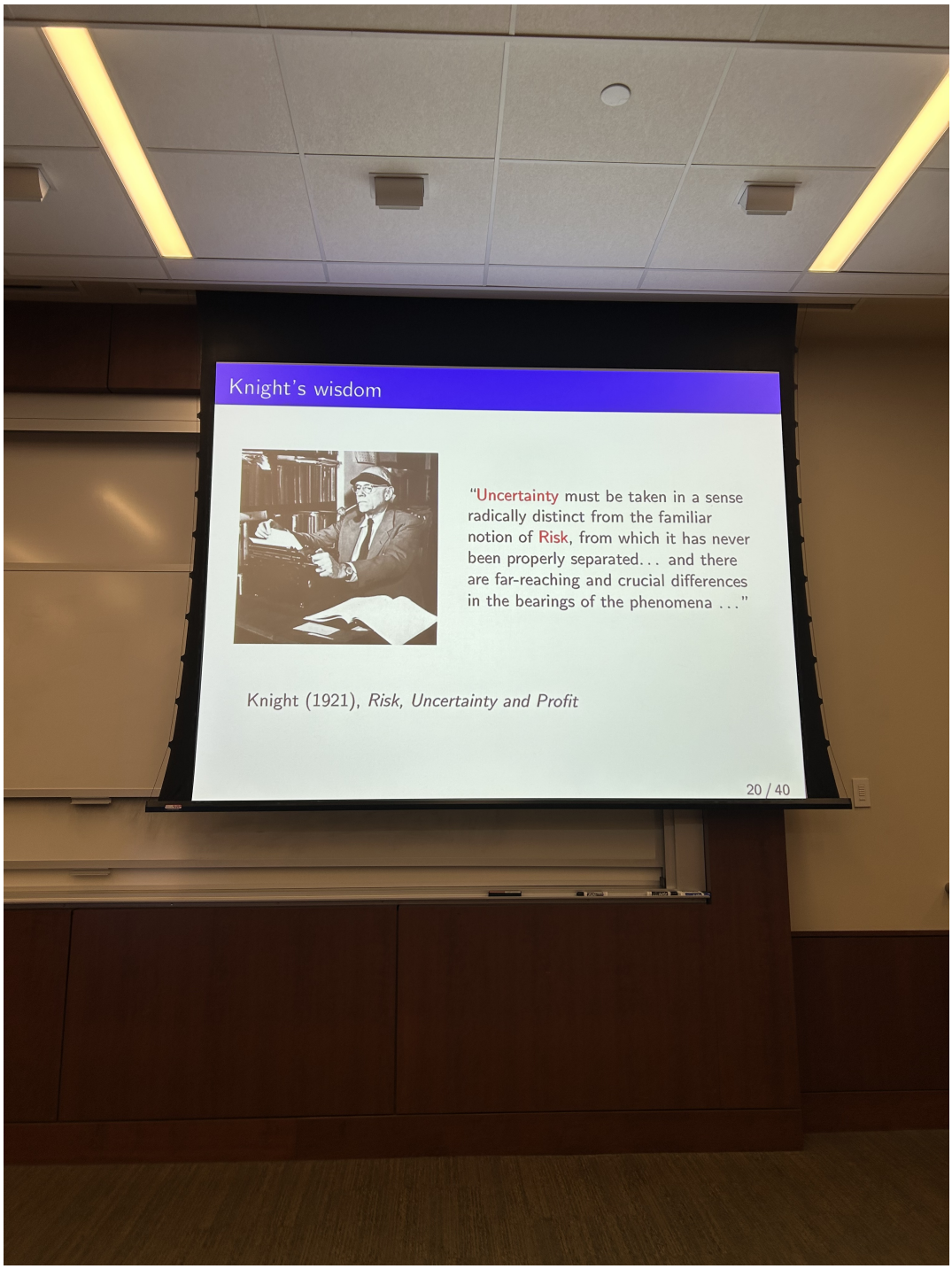


Figure 3: Uncertainty vs risk– a distinction we don't usually discuss in the stat-ML community.

some policymakers will then question their legitimacy.

Outcomes but probabilities are known: risk

Which among multiple probability models is best: ambiguity.

Ways in which a model might give flawed probability predictions: misspecification.

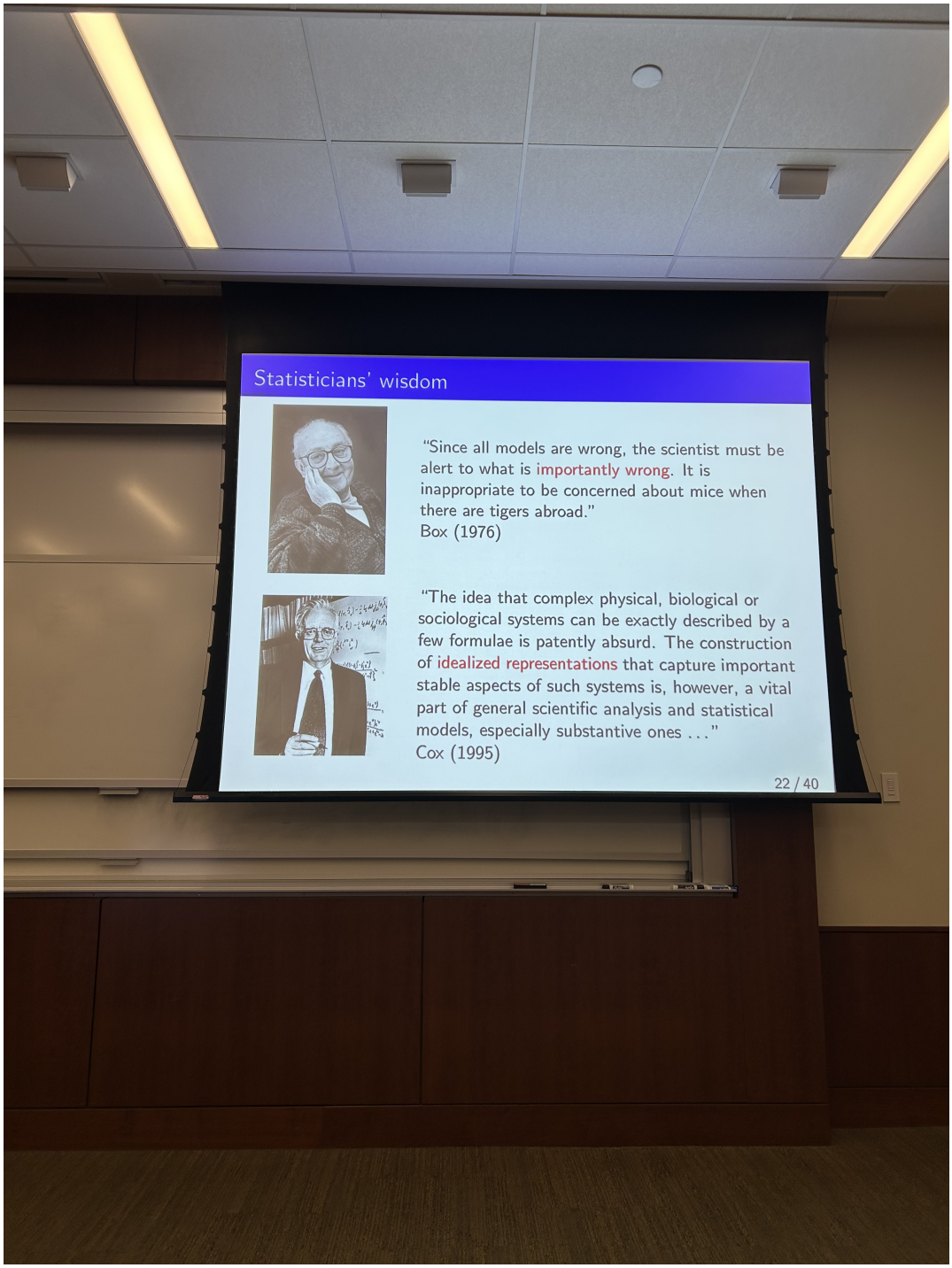


Figure 4: Seminar slide.

[SJ: Make sure I understand the distinctions.]

"We endow our policymaker with preferences that encode these uncertainty aversions."

[SJ: My guess: it basically becomes a max min robust optimal control problem, as he hinted about above by referring to investment and emission decisions as maximizing controls, and the probability model as a minimizing variable? To confirm.]

[SJ: Q from audience: smooth ambiguity, not Gilboa–Schmeidler (?).]

Basically a sensitivity analysis [SJ: ?].

Misspecification aversion: divergence wrt a baseline model. [SJ: Very arbitrary.]

Ambiguity aversion: divergence wrt subjective priors/posteriors

We invoke the min-max theorem, we investigate the worst-case prior. [SJ: Analogous to the least-favorable prior from minimax theory in statistics.] This distribution identifies the "most concerning" "types" of misspecification. We then use the worst-case adjusted probability to compute valuations.

[SJ: A very skeptical Q from the audience: what is the input? Why is the input reasonable? Reply is clear.]

[SJ: Had to step outside at slide 27/40, came back at slide 29/40; ask Prof Hansen for slides 27 and 28 after the talk; is it just the robust min max problem?]

Part III:

AK model [SJ: ?] for climate change-related "big project" R&D investment and "technology discovery", and discuss investment implications.

Ex: nuclear fusion investment.

The technology uncertainty, with respect to economically viable technologies, dominates the uncertainty concern in this model.

See plot of uncertainty-adjusted probabilities for the technology jump only specification; as you crank up aversion wrt the number of years till a technological breakthrough (20, 40, 60, ...), you see that the worst-case prior shifts. (In the model, "arrival of (the first) technological jump" is modeled as the first jump for a particular type ℓ.)

Plot of investment as a function of the inverse ξ_m^{-1} of the "robustness penalty parameter" [SJ: appears in the formula in the min-max problem I missed?]. It's

That's non-intuitive; why isn't it monotone?

The punchline (!) is that there's a tradeoff, a fight between two terms in a product, as illustrated in the following slide. [SJ: I haven't digested it yet.]

Follow-up work on mean-field control, mean-field games. Also computational challenges to

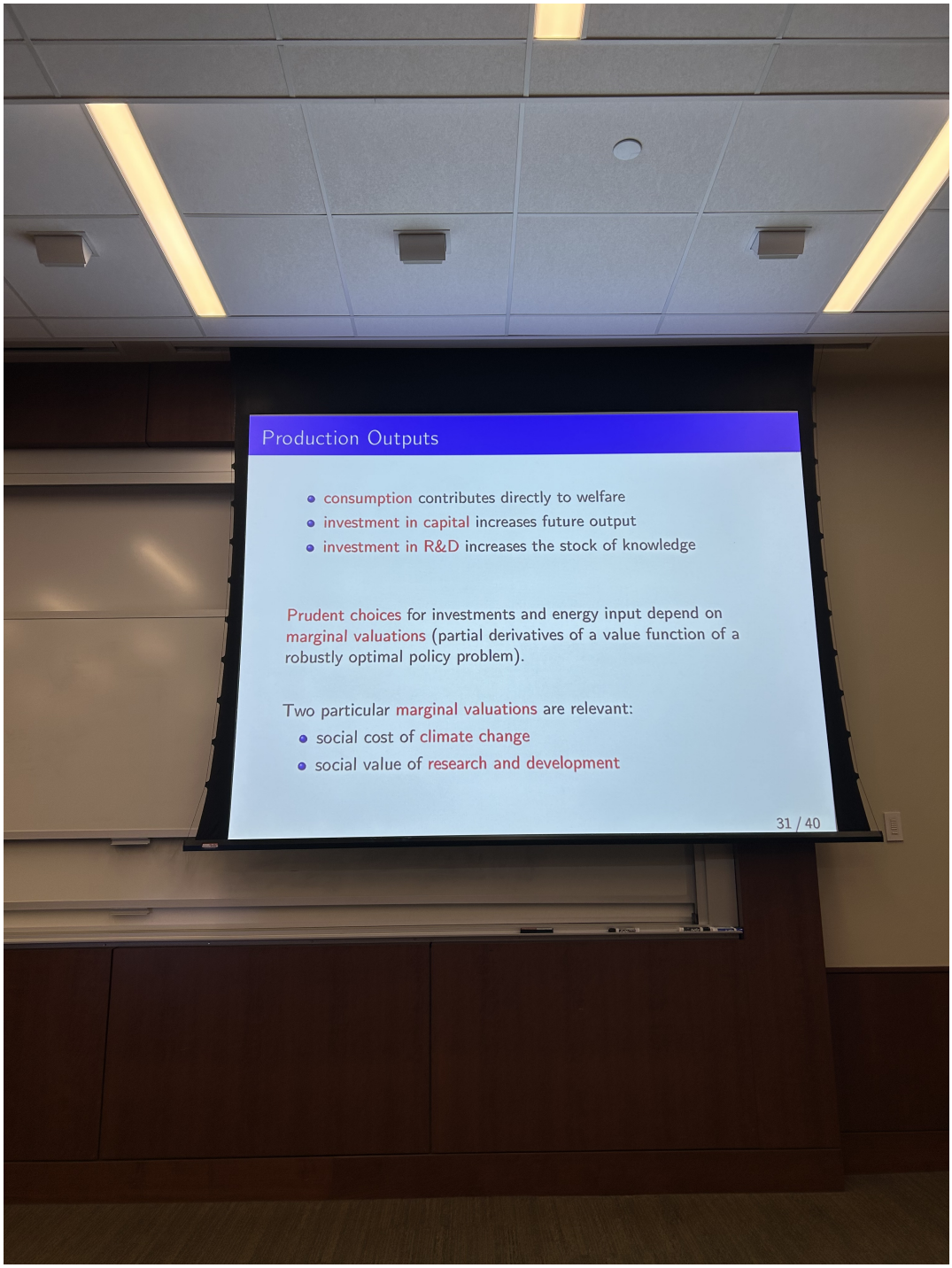


Figure 5: Seminar slide.

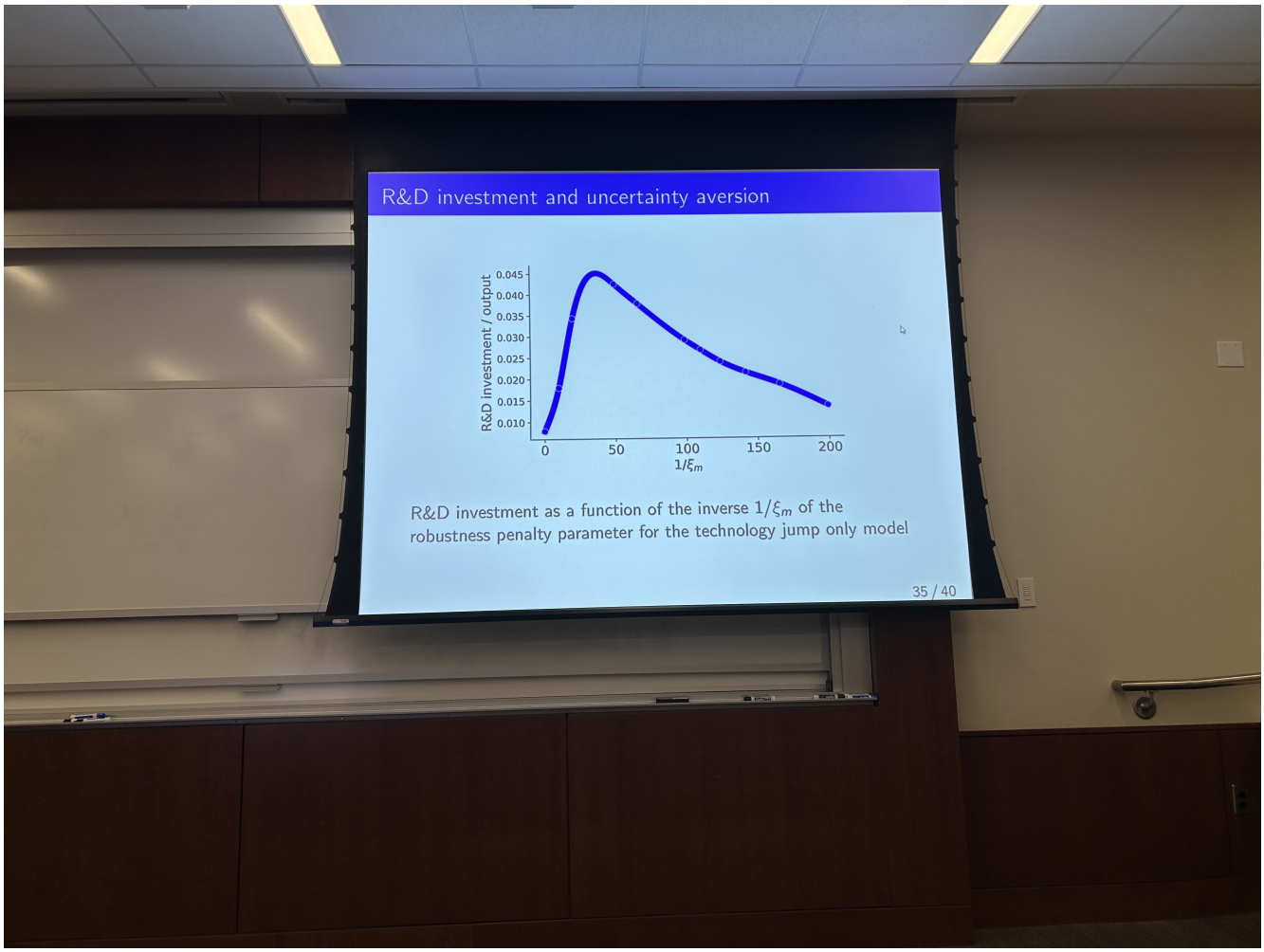


Figure 6: Seminar slide.

solving. [SJ: !]

[SJ: My Q (because I missed slides 27 and 28): what's the formula for worst-case model? Reply: he's doing KL-regularization, hence it ends up being an exponential tilt of the original measure; you could consider other natural divergences, like f-divergences, etc.]

[SJ: My next question: what's the difference between mean-field control and mean-field games? His reply: MFC "couples" the participants in such a way that you just solve a single optimization problem, as opposed to solving a game. It's computationally easier.]

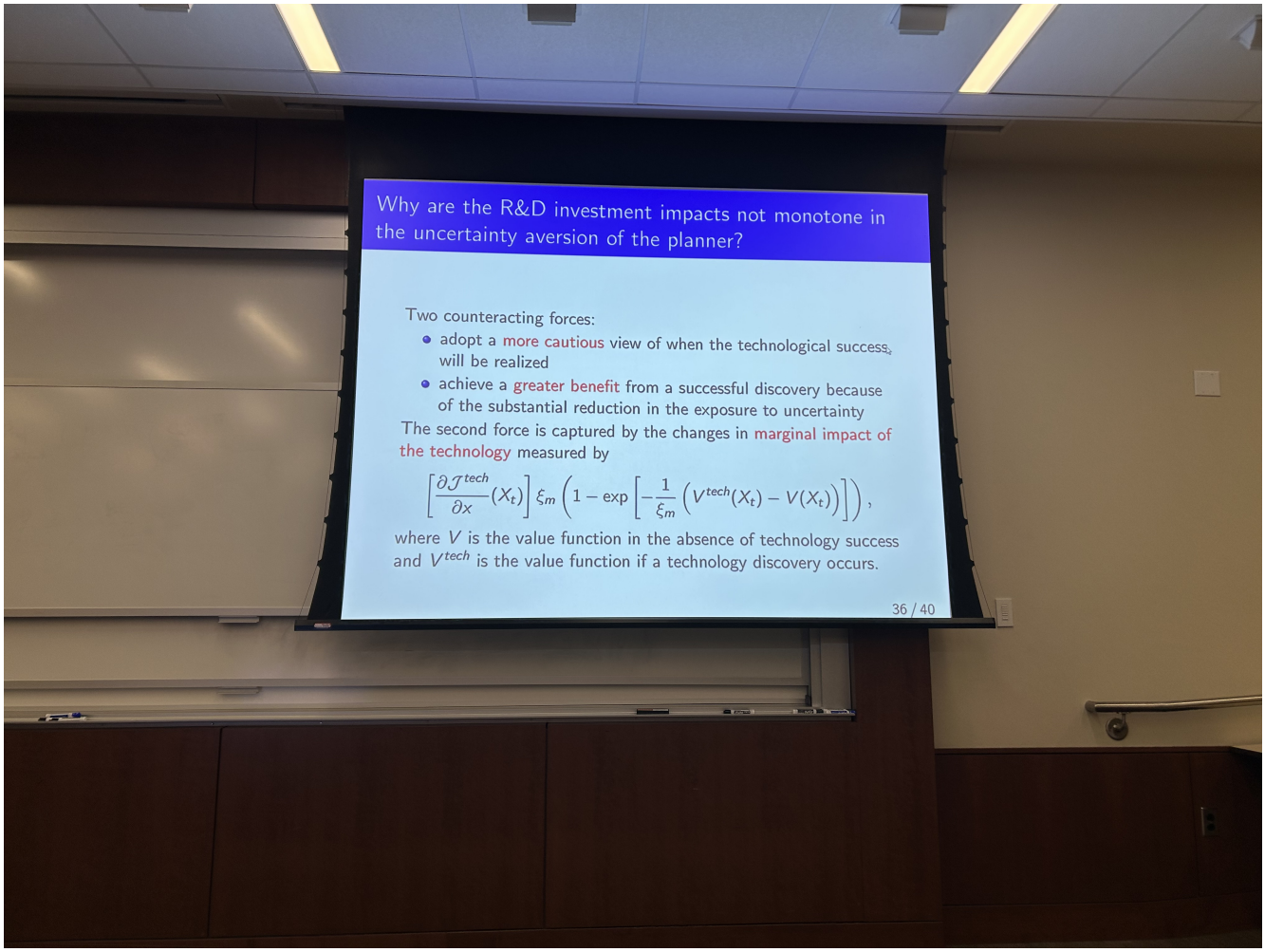


Figure 7: Seminar slide.

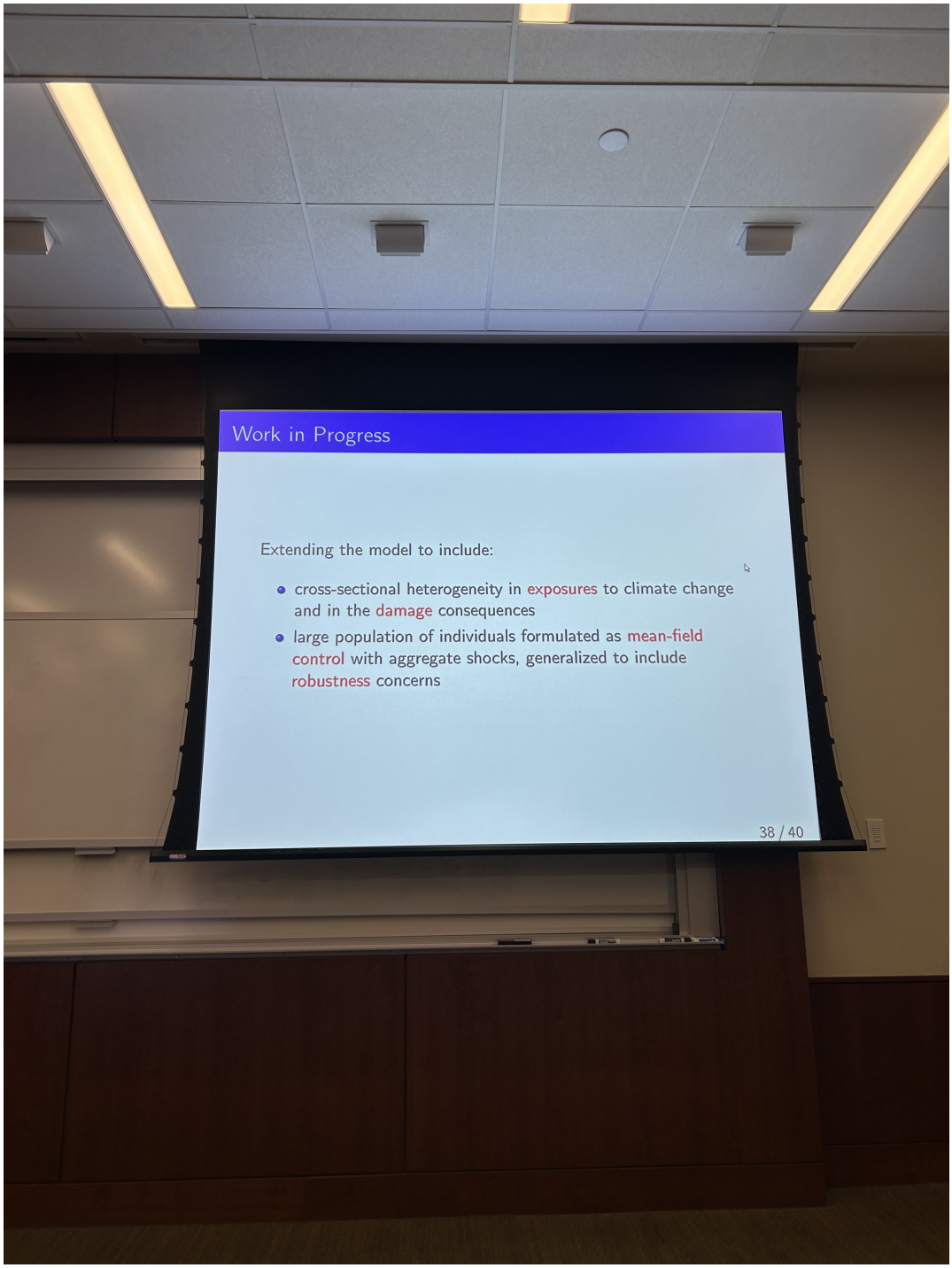


Figure 8: Seminar slide.